
Convolutional Networks (CNN)

Pay attention to the following points when performing the exercises.

1 - Implement computer implementations in Python (PyTorch or TensorFlow library)

Give.

and comparison tables. 2. The results should include graphs of accuracy, error ,

3 - The final report must include numerical analyses and interpretation of results. Submitting only code without analysis

It is worthless.

And save your report in a compressed file with the format zip03.HW_StudentID

4 - Code files

Upload it to the course website by the due date.

should not lead to 5 - The use of artificial intelligence code generation tools is permitted for learning purposes only and

Directly copy the code or analyze it.

Investigating the sensitivity of convolutional networks to basic deep learning concepts such as In this exercise, your goal is to
normalize, initialize, and regularize. All experiments should be performed experimentally and the analyses should be
Numerical results, graphs, and documented results should be included in the final report. For this exercise, the Fashion- dataset was used.
Use MNIST . The py.dataset file attached to the training definition contains sample code for
Each experiment should be repeated at least three times to ensure stability.Loading and working with this dataset will display the results
to be analyzed.

Question 1

Review of Normalization Methods

In this section, we examine the impact of different normalization methods on the learning process. The goal is to

And the features improve the convergence or sustainability of education. Let's see how data normalization works.

1. Design your basic network. This network should consist of at least two convolutional layers, one MaxPooling layer, and a final Fully Connected layer. The network structure should be kept constant throughout all experiments.

2 - Execute the following three modes:

1 - Without any normalization (just dividing the pixel values by 255)

2 - With batch normalization (after each convolution layer)

3 - With a Normalization Layer (after each convolution layer)

And plot the training and validation errors and analyze the convergence behavior. 3 - For each case, the accuracy graph Show.

4 - Explain which method has the most stability and the least fluctuation in education and why?

Question 2

Analysis of the impact of initialization

weights on learning quality is examined. In this section, the effect of different methods of initializing

1 - Train the same basic network with three different initializing methods:

Random Normal Initialization - 1

Xavier (Glorot) Initialization - 2

He Initialization - 3

2 - The learning rate and other settings must be constant in all three cases so that the results are comparable.

3 - Draw a graph of the change in training and validation accuracy and explain which method converged faster and which achieved a higher accuracy value.

4. Given the mathematical relationships in the original article for each method, analyze why the observed behavior makes sense. It is.

Question 3

Analysis of the effect of regularization

Investigating the effect of different regularization methods in reducing overfitting .

In this section, the goal

1. Perform four separate experiments :

1 - Without regularization

2 - With Regularization 2L (value 0.001)

0.5 and 3 - With Dropout at rates of 0.3

4 - By applying Augmentation Data including rotation, displacement, and horizontal inversion

2 - Present the results of each case in the form of training/validation graphs and confusion matrices.

3 - Draw comparative graphs and tables of the experiments and explain based on the results of the experiments which

And which one has slowed down learning? Do you think?

The method has better control overfitting. What is

the reason? Answer using the results of the experiments.

How do the results change if two or more methods are combined (e.g. Dropout + 2L)? 3 Combinations 4 -

tested different ones and after drawing comparative graphs and tables, based on the experiments conducted,

Answer.

Question 4

Summary and sensitivity analysis

1 - Based on all the tests performed, the most sensitive network component to configuration changes is

Specify (normalization, initialization, or regularization?)

2 - Present your results in a summary table of the best settings along with the final accuracy and convergence speed.

Report.

3 - In a few paragraphs, present your final analysis from an engineering and mathematical perspective.